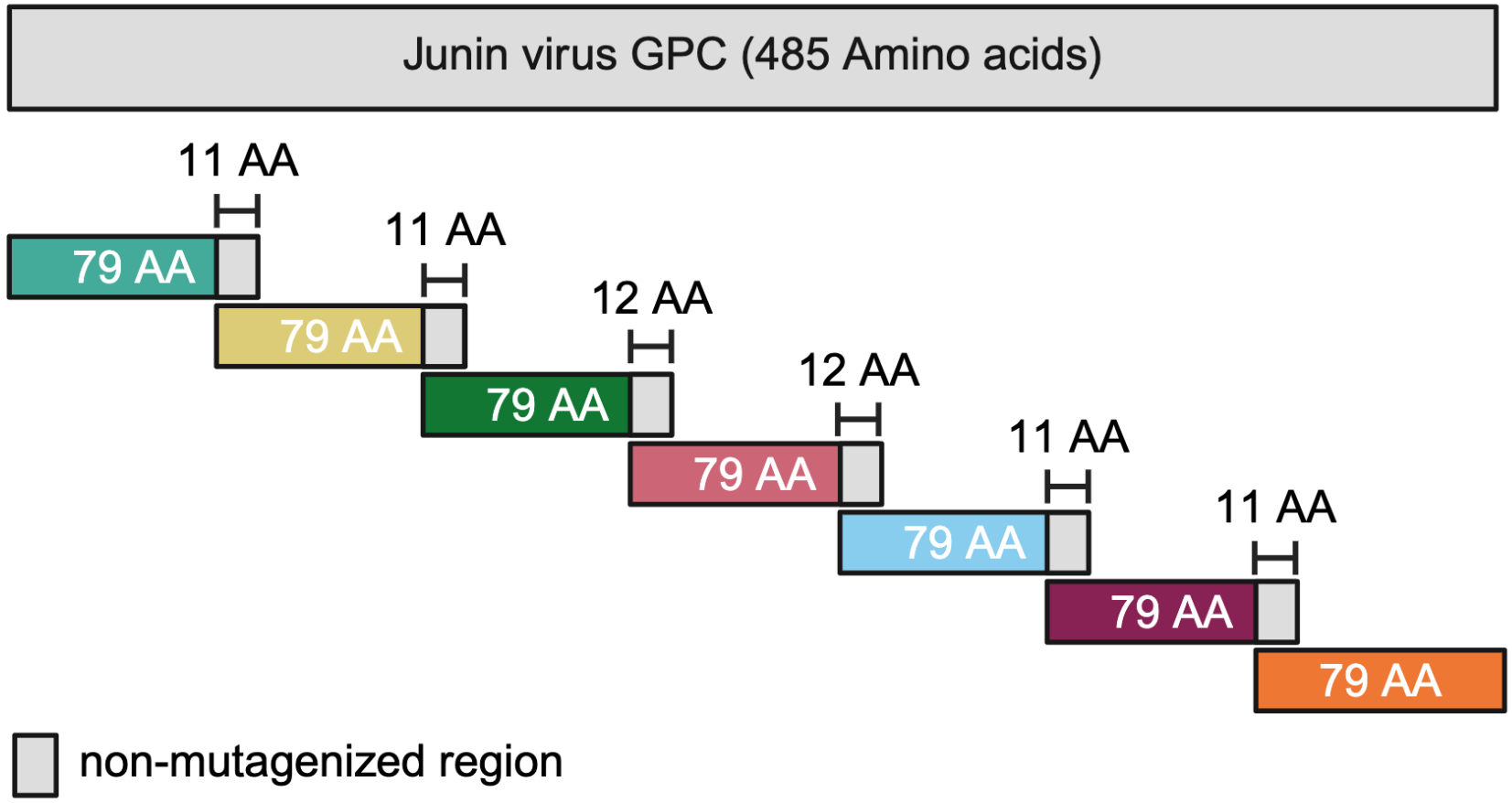
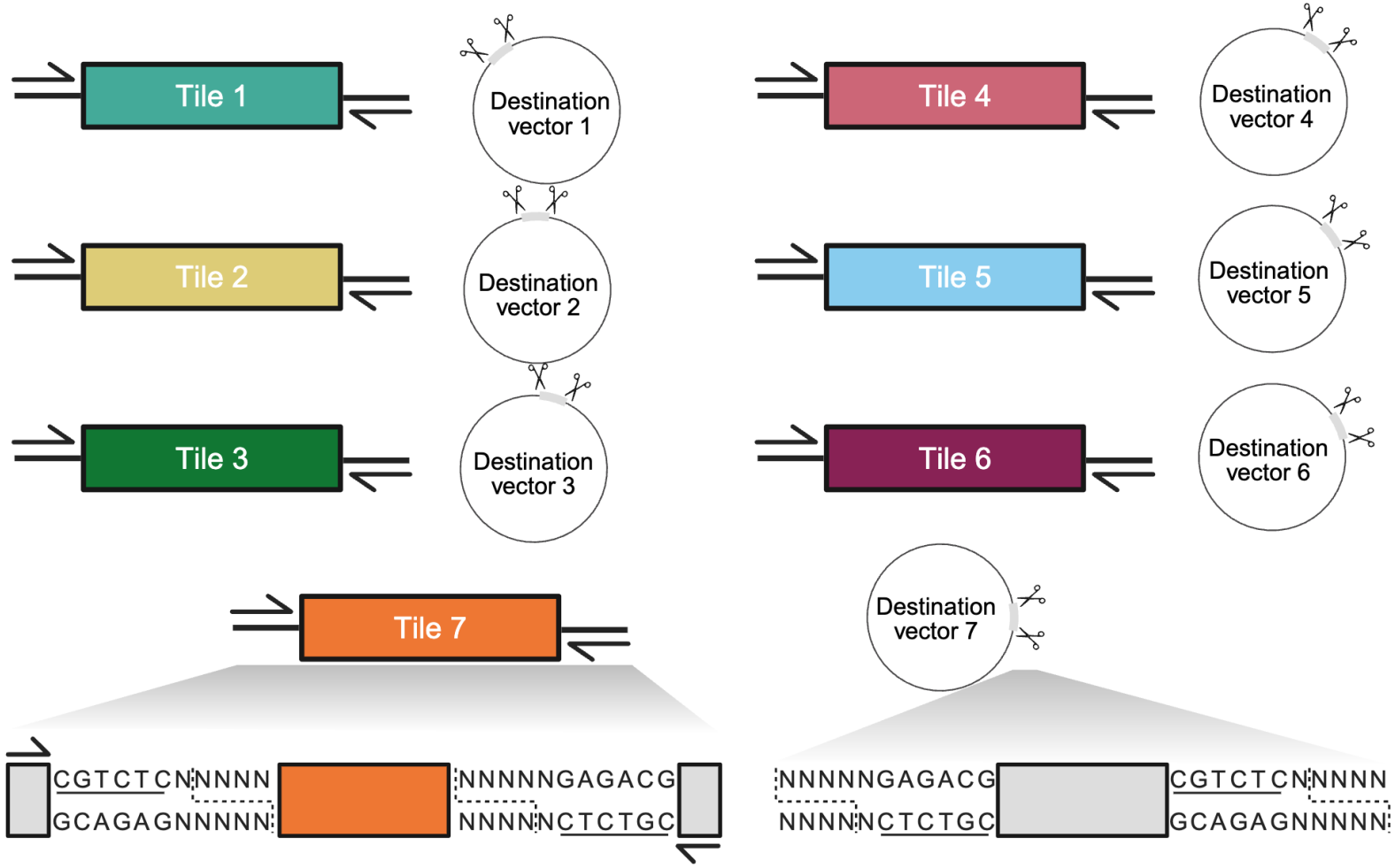
Junin virus GPC deep mutational scanning library construction

Mutagenized tiles



- **Stop codons are designed every 3rd position for the first 100 sites (33 stop codon variants)**
- **19 alternative AAs * 485 = 9,215 variants (+33 stop codon variants) = 9,248 variants**
- **7 wildtype oligos included for tiles 1-6 and 8 wildtype oligos included for tile 7**

Tile and destination vector design



Overhang specificities

Tile 1

	CACC	GGTG	CTGT	ACAG
CACC		■		
GGTG	■			
CTGT				■
ACAG			■	

Tile 2

	CACC	GGTG	TTCA	TGAA
CACC		■		
GGTG	■			
TTCA				■
TGAA			■	

Tile 3

	AAGG	CCTT	CAGC	GCTG
AAGG		■		
CCTT	■			
CAGC				■
GCTG			■	

Tile 4

	AGTA	TACT	AAGT	ACTT
AGTA		■		
TACT	■			
AAGT				■
ACTT			■	

*1bp spacer to move overhang

Tile 5

	CTAC	GTAG	CTGA	TCAG
CTAC		■		
GTAG	■			
CTGA				■
TCAG			■	

Tile 6

	ACCC	GGGT	CAGC	GCTG
ACCC		■		
GGGT	■			
CAGC				■
GCTG			■	

Tile 7

	TAAT	ATTA	GATG	CATC
TAAT		■		
ATTA	■			
GATG				■
CATC			■	